Task 3 — using the earlier system to precondition the later one

Sys 1 (monodomain) $\to$ Sys 2 (recover u_e, singular): implement all options, find the best

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`asm_bug_demo/cardiac/xsys_precond.c`

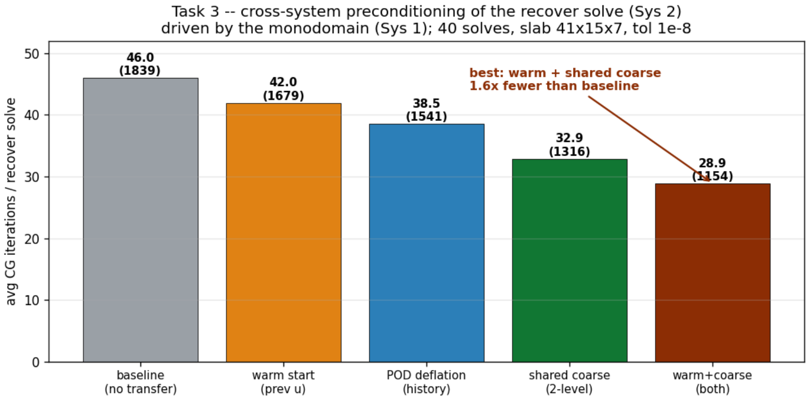
Problem definition — the exact example

the recover solve driven by the monodomain, every time step

Sys 2 : $\quad K_{\sigma_i+\sigma_e}\, u_e = -K_{\sigma_i} V_m(t) \qquad$ (pure Neumann, SINGULAR).

- As the depolarization wave sweeps the slab, the RHS and $u_e(t)$ vary **smoothly in time** $\Rightarrow$ the solution sequence lives near a low-dimensional subspace.
- Example: Niederer slab $20{\times}7{\times}3$ mm, $h{=}0.5$ (4305 nodes); $\sigma_i{=}(0.17, 0.019)$, $\sigma_e{=}(0.62, 0.236)$ S/m; a moving sigmoid front (0.6 m/s), 40 consistent (mean-zero) RHS.
- Metric: CG iterations per recover solve to $\|r\|/\|b\| < 10^{-8}$.

All cross-system options, and the best



- **Shared coarse space is the main lever** (46→33, 1.4×): the Nicolaides coarse space built once on the shared heart mesh spans the constant = the singular Neumann null space = the slow global mode (Task 1's result) ⇒ removes the singularity *and* lifts $\lambda_{\min}$.
- **Solution history (warm/POD) is secondary** (the moving front is high-rank in solution space ⇒ limited POD benefit, the classic advection limitation).
- **Best = warm + shared coarse** (orthogonal: time-coherence + low-end): **28.9 avg, 1.6× fewer** than baseline.

Conclusion (Task 3)

- **Best cross-system preconditioner = shared Nicolaides coarse space (Sys 1 $\to$ Sys 2 structure transfer) + history warm-start.** $1.6\times$ fewer iterations here.
- The **coarse space** is the dominant, most-transferable piece: same mesh $\Rightarrow$ build once, reuse; it *exactly* cancels Sys 2's singular constant mode (Task 1).
- **History** (warm/POD) adds time-coherence but is limited by the moving front (advection $\Rightarrow$ high-rank); it dominates only for slow fronts / loose tolerance.
- **Operator-level transfer** (use Sys 1's $\frac{1}{\Delta t}M + \frac{1}{2}K$ as Sys 2's preconditioner) is *not* the winner: the mass shift makes Sys 1's spectrum mismatch the pure-Laplacian Sys 2.